WEAPON DETECTION SYSTEM

Dataset Engineering, Model Development, Evaluation and Deployment

FINAL PROJECT REPORT

Final implementation status: Complete

Report date: 27 September 2026

Primary model: V3.pt (YOLO26m, 1280 px)

Frontend: Streamlit | Image • Video • Live CCTV

Project closure
Model development, dataset engineering, controlled evaluation, and frontend integration have been completed. The final system is documented as a research/portfolio proof of concept rather than a production or safety-critical security product.

Executive Summary

This project developed an end-to-end weapon detection system aimed at realistic surveillance-style imagery and deployable through a simple visual interface. The work began as a dataset-engineering problem and evolved into a complete computer-vision application covering image analysis, uploaded-video review, and live CCTV monitoring. The final detector recognizes five classes: Knife, Machette, Handgun, Longgun, and Other.

A major focus was preventing dataset leakage and avoiding the common failure mode of simply increasing frame counts. The project preserved source videos, froze the class mapping before annotation, split surveillance data at the video level, audited label integrity, and used visual redundancy analysis to reduce repetitive frames. The resulting verified master dataset contains 15,288 images, 14,129 annotated objects, and 3,784 empty-label images, with a final 12,199/2,305/784 Train/Validation/Test split and zero source-sequence leakage.

Model development progressed through V1, a hard-example-heavy first V2 experiment, V2.1, V2.2, and V3. The first V2 experiment failed because it replaced too much of the broad V1 training distribution with a smaller failure-focused pool. V2.1 corrected this by preserving the full V1 training distribution and adding 1,436 unique hard examples. V2.2 then added a targeted 354-image aerial adaptation set. V3 was initialized from V2.1 and trained on the 13,989-image V2.2 training set at 1280 px.

The final evidence is intentionally domain-specific. On the original locked 784-image Test set, the apples-to-apples 1280 px evaluation recorded V2.1 at precision 0.678, recall 0.628, mAP50 0.601 and mAP50-95 0.413, while V3 recorded 0.658, 0.544, 0.538 and 0.356. On the separate held-out aerial test of 380 frames from 38 unseen clips, V3 full-frame inference recorded precision 0.830, recall 0.437, F1 0.572, specificity 0.911 and FPR 0.089. This result is complemented by clip-level precision 0.765, recall 0.684 and F1 0.722.

The final application uses V3.pt as the primary weapon detector. Person detection is retained as an optional advanced path rather than the default. Video review groups detections into time ranges and allows the user to jump to a selected range without autoplay. Live CCTV groups consecutive detections into incident tiles and preserves the current session after stopping; clearing the current session removes those prior tiles, and a newly started session begins clean.

Final outcome

The project is complete as a documented research/portfolio proof of concept. The model-development phase is frozen, and the final emphasis is reproducible use of the frontend and clear presentation of the evidence.

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1. Introduction and Problem Definition

1.1 Background

Weapon detection in surveillance imagery is fundamentally different from classifying clean product photographs. Weapons may appear at small scales, be partially occluded by people, be blurred by motion, be viewed from unusual angles, or visually resemble ordinary objects. The project therefore treated the problem as an object-detection and deployment problem rather than a simple image-classification task.

The practical objective was to build a reusable detection core that can identify visible weapon instances and then expose that capability through three user-facing input modes: still images, uploaded video, and live CCTV/stream sources. A further goal was to study how the detector behaves under domain shift, especially when surveillance imagery is visually different from the main training sources.

1.2 Problem Definition

Given an input image or video frame, the system must determine whether visible objects correspond to one or more of the five project weapon classes, localize those objects with bounding boxes, provide confidence scores, and produce a human-reviewable output. For video and live CCTV, individual detections should also be interpreted over time so that repeated detections can be grouped into meaningful events rather than presented as a flood of near-identical alerts.

1.3 Design Philosophy

- Weapon-first inference is the primary operating mode. Person-aware analysis is optional and exists for difficult scenes or diagnostic experiments.
- Dataset quality and leakage control are treated as first-class engineering tasks, not post-processing details.
- Difficult examples should be mined selectively rather than allowed to replace broad training coverage.
- Official benchmark metrics and supplemental failure-analysis metrics are reported separately.
- The final application should expose evidence clearly enough for a human to verify what the detector saw.

1.4 Research Questions

1. How can heterogeneous weapon datasets and surveillance frames be merged without class-ID drift or source leakage?
2. How does targeted hard-example mining affect detector generalization?
3. Does adding a small, clip-level-selected aerial adaptation set improve held-out aerial behavior?
4. How can a detection model be converted into a usable image, video and live-CCTV application without obscuring model limitations?

2. Project Objectives and Requirements

2.1 Objectives

- Construct a verified YOLO-compatible dataset with a frozen five-class taxonomy.
- Use surveillance video as a domain-alignment component instead of relying only on isolated weapon imagery.
- Prevent temporal leakage by keeping source videos and source clips within a single split.
- Develop, compare and document multiple model iterations rather than accepting the first trained checkpoint.
- Analyze failure modes, especially small-object misses, hard negatives, class confusions and domain shift.
- Provide a practical frontend for image, video and live CCTV analysis.

2.2 Functional Requirements

Requirement	Final implementation
Image input	Upload JPG/JPEG/PNG; run weapon analysis; show original and annotated result.
Video input	Upload MP4/MOV/AVI/MKV; sample frames; retain analyzed result; show grouped detection ranges.
Video navigation	Detection range/timestamp controls seek the embedded video while keeping playback paused.
CCTV input	Webcam index or RTSP/HTTP/RTMP source; latest-frame processing; incident grouping and evidence tiles.
Advanced analysis	Optional person-aware path using person detection, ROI processing, tracking and temporal confirmation.
Evidence	Store/retain incident snapshots and show peak-confidence evidence rather than only the first detection.
Session handling	Stop preserves current session; Clear Current Session clears it; a new session starts empty.

2.3 Non-Functional Requirements

- Run within practical consumer/GPU and Colab resource limits.
- Keep original datasets and validated artifacts recoverable.
- Use clear, interpretable UI terminology such as class score, confidence, and detection status.
- Fail safely when a region is too small or analysis is inconclusive rather than inventing a weapon label.

3. Data Sources and Dataset Strategy

The project combined four main data streams: Dangerous Items, HGP, a Kaggle weapon dataset, and a project-collected surveillance video corpus. The sources served different purposes. Existing datasets supplied broad class diversity, while the surveillance videos supplied realistic scene structure and temporal variation.

3.1 Source Contributions

Source	Observed inventory	Role
Dangerous Items	10,140 images/labels; 9,144 annotations; 2,198 empty labels	Large processed source; required class-ID normalization.
HGP	2,199 images/labels; 2,866 annotations	Processed handgun source; retained as unified Handgun.
Kaggle	713 images/labels; 1,152 annotations	Supplementary weapon-appearance diversity; IDs already aligned after audit.
Surveillance corpus	85 videos; 3,197 extracted frames	Domain-focused source for realistic CCTV-style imagery.

3.2 Frozen Taxonomy

Unified ID	Class	Definition
0	Knife	Knives and knife-like target instances.
1	Machette	Machetes and comparable blade forms under the project taxonomy.
2	Handgun	Pistols and handguns.
3	Longgun	Rifles, shotguns and other long firearms.
4	Other	Dangerous items outside the main four classes when in-scope under project policy.

Important data-quality decision
Class names were never assumed to share the same numeric IDs across sources. Explicit source-to-unified mappings were audited before merge. The mapping was frozen before annotation and preserved

throughout model development.

3.3 Why the Taxonomy Includes Other

The source data contains visually diverse objects that do not fit cleanly into Knife, Machette, Handgun or Longgun. Retaining Other avoids forcing every source object into an incorrect specific class. At the same time, Other is not a label for annotation uncertainty; genuinely ambiguous objects can remain unlabeled.

4. Surveillance Video Preparation and Frame Engineering

4.1 Source Video Corpus

The surveillance collection contained 85 source videos: H001–H034 for Handgun, K001–K027 for Knife, L001–L019 for Longgun, and M001–M005 for Machette. After extraction, the corpus contained 3,197 frames: 1,135 Handgun, 1,270 Knife, 596 Longgun and 196 Machette.

4.2 Video Preparation

- Irrelevant material was trimmed while preserving surveillance characteristics.
- Intrusive overlays were handled where they interfered with the scene.
- Natural CCTV timestamps were retained when non-intrusive because they are part of the target domain.
- Viewpoint, distance, angle and environment variation were preserved instead of converting footage into product-style imagery.

4.3 Frame Integrity

The frame hierarchy was initially misinterpreted because the actual structure contained an additional source-video folder beneath each category. A dedicated integrity pass was then used to check complete frame-number sequences, duplicate numbers, missing frames and naming consistency. The final result was 85 video folders, 3,197 frames and zero detected integrity issues.

4.4 Preventing Temporal Leakage

Random frame-level splitting was explicitly rejected. Adjacent frames from a single source video are strongly correlated, so placing neighboring frames into Train and Test can produce an overly optimistic estimate. The surveillance corpus was therefore split at the source-video level, with category representation maintained across the three splits.

Split	Videos	Frames
Train	59	2,236
Validation	13	438
Test	13	523

4.5 Frame Selection and Redundancy Analysis

A first temporal candidate pool contained 1,150 frames. A visual-diversity pass used perceptual hashing and normalized pixel difference; 494 of 1,065 candidate comparisons were highly similar (46.38%).

Manual inspection showed that some similar-looking pairs still contained meaningful hand or weapon movement, so similarity was not used as a blunt delete rule.

The final strategy grouped consecutive similar candidates into redundancy runs: single frames were retained, two-frame runs kept both frames, and longer runs retained the first, middle and last frames. An initial adaptive pass produced 848 candidates, but the rare Machette Test video was reduced from 15 to 4 candidates. To protect the unseen test set, all 184 Test candidates were preserved. The final locked selection contained 890 candidate frames: Train 589, Validation 117, Test 184.

5. Annotation, Quality Control and Dataset Assembly

5.1 Annotation Environment

Annotation was performed locally with CVAT v2.73.0 running through WSL2/Ubuntu and Docker. A dedicated Weapon_Detection project was created with the frozen five-class mapping. Tracking tools were used on some continuous sequences to reduce repetitive box drawing, with explicit termination when an object left the frame.

5.2 Annotation Rules

- Bounding boxes cover the weapon itself, not the person, hands, clothing or background.
- Boxes are reasonably tight while retaining all visibly identifiable weapon parts.
- Each visible weapon receives its own box.
- Partially occluded or cropped weapons are annotated when enough of the object is visibly identifiable.
- Invisible weapons are not guessed from neighboring frames or context.
- Natural overlays such as timestamps and logos are ignored as target objects.
- Empty labels are retained when a selected image contains no confidently identifiable weapon.

5.3 Pilot Annotation and Full Conversion

A 213-image pilot batch was first annotated and quality-checked to establish the working rules. The pilot contained approximately 76 Handgun images, 75 Knife images, 54 Longgun images and 14 Machette images. The final annotation workflow then covered the 2,236-image surveillance training-side pool. Five CVAT XML exports were converted into YOLO-compatible labels and validated.

Validation result	Count
Images	2,236
YOLO label files	2,236
Object annotations	967
Empty labels	1,421
Missing image-label pairs	0

5.4 External Dataset Normalization

Dangerous Items required recursive normalization into the frozen taxonomy. HGP used its processed Handgun class directly. Kaggle class IDs 0, 2, 3 and 4 aligned with Knife, Handgun, Longgun and Other after audit. The normalized sources were validated independently before merge.

5.5 Master Dataset Merge

Four verified sources were merged into a new master directory: CVAT_YOLO_Converted, Kaggle, HGP and Dangerous_Items_Normalized. Globally unique filenames were generated and a merge manifest preserved source traceability. An independent audit confirmed 15,288 images, 15,288 labels, 14,129 annotations, 3,784 empty labels, zero missing image-label pairs, zero duplicate stems, zero malformed labels, zero invalid boxes and zero unexpected class IDs.

5.6 Final Internal Split

Several candidate split planners were tested and rejected when they produced poor source or class balance despite zero leakage. The final approved split used 80% Train, 15% Validation and 5% internal Test while preserving source-sequence integrity. The final counts were 12,199 Train images, 2,305 Validation images and 784 Test images.

Split	Images	Objects	Class distribution
Train	12,199	11,119	Knife 2,002 | Machete 1,412 | Handgun 2,711 | Longgun 3,331 | Other 1,663
Validation	2,305	2,235	Knife 365 | Machete 257 | Handgun 495 | Longgun 813 | Other 305
Test	784	775	Knife 124 | Machete 87 | Handgun 176 | Longgun 286 | Other 102

6. Model Development and Experiment History

6.1 Training Environment

Setting	Value
GPU	NVIDIA Tesla T4
Available VRAM	14.56 GB
Python	3.13.x in later runs
PyTorch	2.11.0 + CUDA 12.8
Ultralytics	8.4.123
Model family	YOLO26m
Primary baseline resolution	640 × 640
Later production resolution	1280 × 1280
Training approach	Transfer learning / checkpoint continuation with on-the-fly augmentation

6.2 Why YOLO26m

The pre-training data analysis showed a sufficiently large image pool to justify a medium-capacity detector. Object-size analysis did not support jumping immediately to extreme resolution for the baseline, so 640 px was used for the initial controlled experiment. Higher resolution was introduced later when external resolution tests showed a clear small-object benefit.

6.3 Development Path

Stage	Training composition / setting	Outcome
V1	15,288-image verified dataset; Epoch-46 frozen checkpoint	Historical foundation; official Test benchmark preserved.
First V2	4,107 representative V1 images + 2,851 hard examples = 6,958 images	Severe validation/Test regression; rejected.
V2.1	Full V1 Train distribution (12,199) + 1,436 unique hard additions = 13,635	Retained baseline.
V2.2	V2.1 + 354 aerial hard examples = 13,989 Train; Validation/Test unchanged	Training dataset used for V3.

V3	YOLO26m, 1280 px, initialized from V2.1 and trained on V2.2	Final frontend candidate V3.pt.

6.4 Historical V1 Benchmark

The authoritative historical Epoch-46 V1 benchmark recorded Validation precision 0.716, recall 0.689, mAP50 0.686 and mAP50-95 0.493. On the locked Test set, V1 recorded precision 0.7211, recall 0.5467, mAP50 0.5655 and mAP50-95 0.4111. These figures remain important historical evidence and are kept distinct from later apples-to-apples local comparisons.

6.5 The Failed Hard-Example-Heavy V2

The first V2 experiment intentionally combined a smaller representative base subset with the 2,851-image hard-example pool. The resulting 6,958-image dataset was structurally clean, but validation performance was poor: precision 0.223, recall 0.127, mAP50 0.0806 and mAP50-95 0.0270. A controlled comparison on the same original Validation and Test splits at 1280 px confirmed severe regression. The lesson was that failure-mined images cannot simply replace the broad training distribution.

Key lesson
Hard examples are most useful as targeted additions to a broad training distribution. They should not be treated as uniformly difficult just because the frozen model originally failed on them.

7. Hard-Example Mining and Domain Adaptation

7.1 Mock Attack

Mock Attack contained 5,149 1920×1080 surveillance-style images with XML annotations. Its source classes were mapped to Handgun, Longgun and Knife. V1 was evaluated externally at 1280 px, and temporal deduplication was used to avoid saving repeated near-identical failures. The approved contribution was 441 images: 263 Handgun, 109 Longgun, 46 Knife and 23 true-negative hard false-positive images.

7.2 Dangerous Items Failure Mining

The existing verified master dataset already contained Dangerous Items, so a duplicate download was not needed. At 1280 px, confidence 0.25 and IoU 0.45, object-level analysis found 7,008 true positives, 1,698 false negatives, 2,731 false positives and 438 class confusions. A crucial correction was made: only 109 of the false-positive images were true negatives; 2,059 positive-context cases already contained real weapons. Those positive-context cases were not relabeled as negatives.

The approved Dangerous Items hard-example pool contained 2,167 images: 740 Knife, 550 Machette, 31 Handgun, 494 Longgun, 243 Other and 109 Negatives. A separate 500-image Positive_Context_FP_Selected pool was preserved independently.

7.3 Sohas

The Sohas DetectionNinja export contained 5,859 images and JSON annotations. After correcting the pairing logic for filenames of the form image.jpg.json, the source passed validation with 5,859/5,859 pairs and zero invalid geometry. V1 failure mining at 1280 px produced 397 object-level true positives, 2,459 false negatives, 7,858 false positives and 1,156 class confusions. After conservative selection and sequence deduplication, 700 images were approved: 250 Knife, 250 Handgun and 200 Negatives.

7.4 Consolidation

The source-specific hard-example pools contained 3,308 records in total: 441 Mock Attack, 2,167 approved Dangerous Items and 700 Sohas. Exact image-content hashing identified 405 duplicate groups and removed 457 duplicate records, leaving 2,851 unique physical images. Where identical image files had different annotations, the annotations were merged; 121 images required annotation merging. Final pairing was 2,851 images to 2,851 labels with zero missing pairs.

Source	Approved images
Mock Attack	441
Dangerous Items	2,167
Sohas	700
Total source records	3,308

Unique after hashing	2,851

7.5 Aerial Adaptation Set

The targeted aerial adaptation set was selected at the clip level. The source manifest contained 253 clips split into 177 training, 38 validation and 38 test clips. Only training clips contributed to the 354-image adaptation set: 184 weapon-source images and 170 umbrella/no-weapon images. Of the 184 weapon-source images, 168 received Longgun boxes and 16 remained empty-label images because a weapon could not be reliably identified in the selected frame. The 38 test clips, containing 19 weapon and 19 no-weapon clips, remained completely held out.

8. Final V3 Training

8.1 Training Configuration

Setting	Final V3 configuration
Starting checkpoint	V2.1 retained baseline
Training data	V2.2: 13,989 Train images; 2,305 Validation; 784 original Test
Model	YOLO26m
Image size	1280 × 1280
Hardware	Tesla T4, 14.56 GB VRAM
Completed epochs reported	E9–E14 after true checkpoint resume
Incomplete epoch	E15 at 13%; excluded from metrics
Checkpoint selected	V3.pt

8.2 Runtime and Checkpoint Recovery

The V3 run progressed through E13 before a later continuation encountered a PyTorch Adam/AMP optimizer-state assertion during resume. A fresh-optimizer continuation was tested but was not retained. A true checkpoint resume restored the E8 state and continued from E9 through E14. Epoch 15 reached only 13% before a runtime disconnect, so it was intentionally excluded from all final claims.

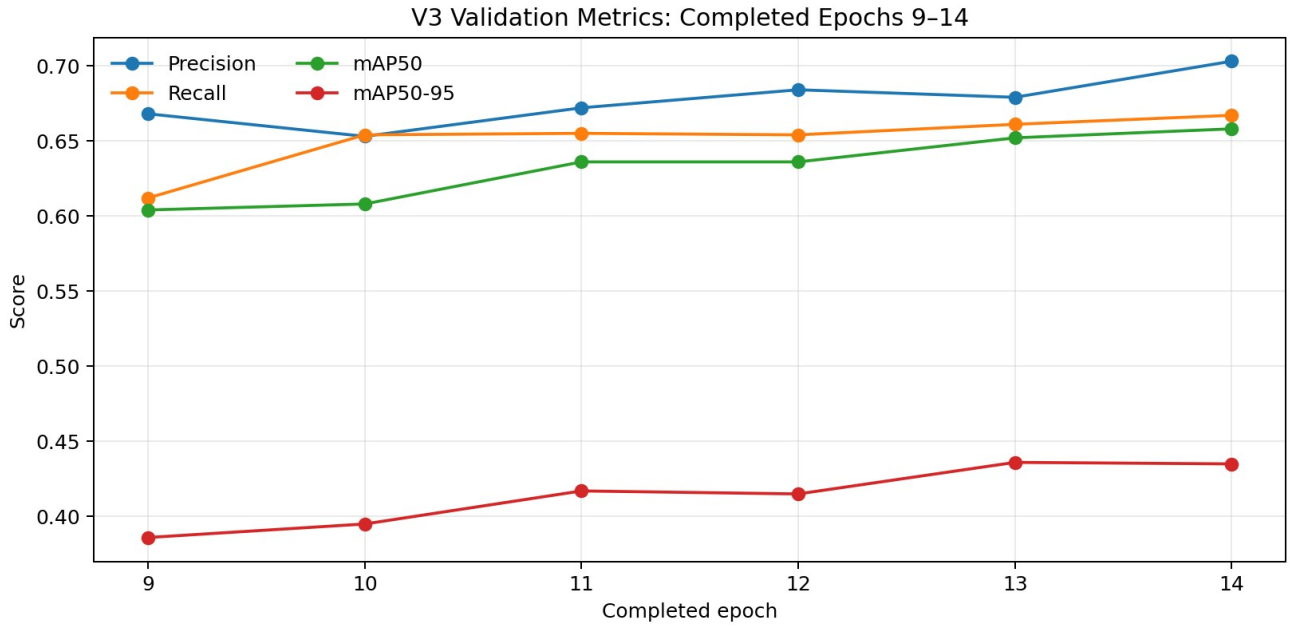


Figure 3. V3 validation metrics for completed epochs 9–14. The reported final state excludes the partial E15 run.

Completed epoch	Precision	Recall	mAP50	mAP50-95
E9	0.668	0.612	0.604	0.386
E10	0.653	0.654	0.608	0.395
E11	0.672	0.655	0.636	0.417
E12	0.684	0.654	0.636	0.415
E13	0.679	0.661	0.652	0.436
E14	0.703	0.667	0.658	0.435

8.3 Final Checkpoint

The completed V3 checkpoint is stored locally as V3.pt and became the deployment model used by the final frontend. The previous retained baseline is preserved locally as V2.pt, which represents V2.1 rather than the rejected first V2 experiment.

9. Evaluation Methodology and Results

9.1 Evaluation Principles

- The original locked 784-image Test set was preserved as a fixed benchmark.
- The final V2.1 versus V3 comparison used the same Test images and 1280 px settings for both models.
- The aerial evaluation used 380 frames from 38 clips that were excluded from V2.2 training.
- Frame-level and clip-level aerial metrics are reported separately because their decision rules differ.
- Person-ROI results are treated as a supplemental diagnostic, not the primary final pipeline.

9.2 Original-Domain V2.1 versus V3

Metric	V2.1	V3	V3 − V2.1
Precision	0.678	0.658	−0.020
Recall	0.628	0.544	−0.084
mAP50	0.601	0.538	−0.063
mAP50-95	0.413	0.356	−0.057

9.3 Class-Level Original-Domain Results

Class	V2.1 P	V3 P	V2.1 mAP50	V3 mAP50	V2.1 mAP50-95	V3 mAP50-95
Knife	0.605	0.604	0.599	0.561	0.409	0.347
Machette	0.568	0.577	0.525	0.493	0.359	0.336
Handgun	0.710	0.731	0.426	0.404	0.306	0.271
Longgun	0.813	0.853	0.696	0.592	0.482	0.419
Other	0.692	0.523	0.757	0.638	0.507	0.408

The original-domain comparison shows why the project does not treat V3 as a universal replacement for V2.1. The intended V3 development target was domain adaptation, especially for aerial imagery, so the original-domain and aerial results answer different questions.

9.4 Held-Out Aerial Frame-Level Results

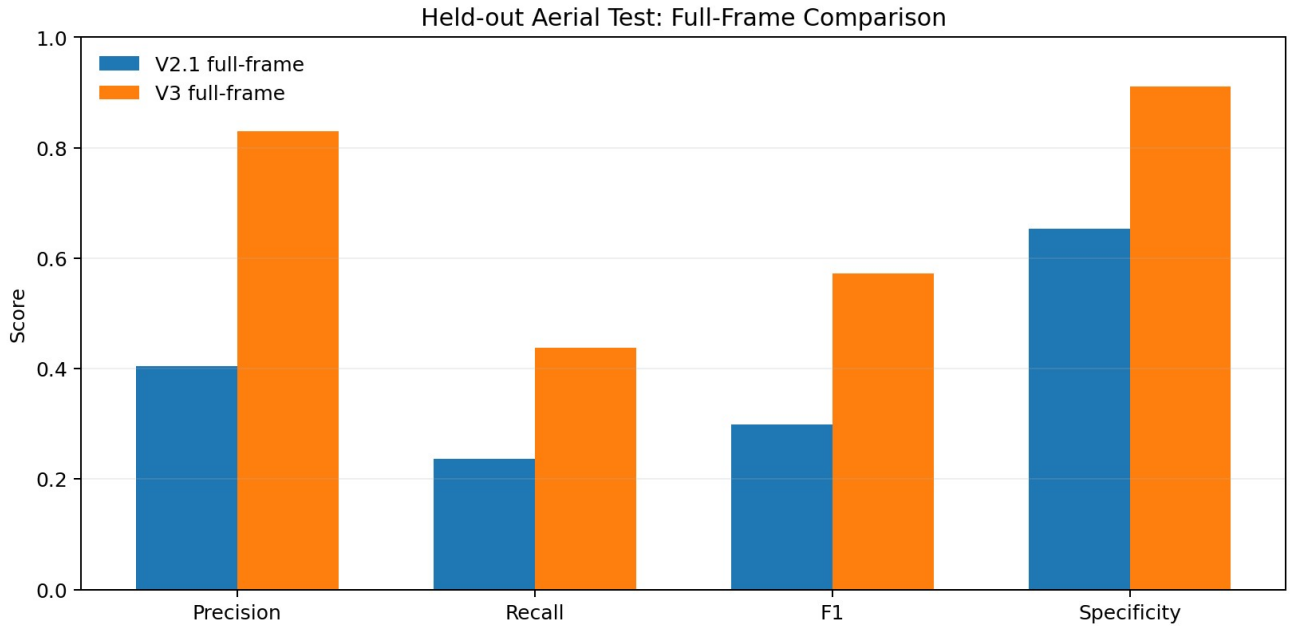


Figure 4. Full-frame comparison on the 380-frame held-out aerial test. Metrics are reported at the frame level.

Metric	V2.1 full-frame	V3 full-frame	Change
TP	45	83	+38
TN	124	173	+49
FP	66	17	−49
FN	145	107	−38
Precision	0.405	0.830	+0.425
Recall	0.237	0.437	+0.200
F1	0.299	0.572	+0.273
Specificity	0.653	0.911	+0.258
FPR	0.347	0.089	−0.258
Longgun recall	0.221	0.437	+0.216

9.5 Held-Out Aerial Clip-Level Results

Clip metric	V2.1	V3

TP	8	13
TN	8	15
FP	11	4
FN	11	6
Precision	0.421	0.765
Recall	0.421	0.684
F1	0.421	0.722
Specificity	0.421	0.789

The clip-level rule treats a weapon clip as detected when at least one frame in the clip fires. Because this differs from frame-level evaluation, clip-level metrics are not interchangeable with frame-level metrics.

9.6 Person-ROI Diagnostic

V3 aerial pipeline	Frame F1	Frame FPR	Longgun recall	Clip F1
Person ROI	0.490	0.247	0.284	0.542
Full frame	0.572	0.089	0.437	0.722

The final test showed that person-ROI processing did not outperform direct full-frame V3 inference for the aerial domain. The observed result suggests that cropping can remove contextual or extended weapon evidence and can also introduce additional false positives. Accordingly, the final primary path remained full-frame V3 at 1280 px.

9.7 Supplemental Broad Aerial Diagnostic

Before the final locked aerial test, a broader 2,530-frame aerial source was used for a supplemental domain-shift diagnostic. Full-frame V2.1 recorded TP 650, TN 445, FP 775 and FN 660, with precision 0.4561, recall 0.4962 and F1 0.4753. The earlier person-ROI V2.1 run recorded TP 910, TN 502, FP 718 and FN 400, with precision 0.5590, recall 0.6947 and F1 0.6195. These results are retained as diagnostic evidence only; the final comparison is the 380-frame held-out test.

10. Small-Object and Resolution Analysis

10.1 Pre-Training Object-Scale Analysis

Metric	Observed
Total images	15,288
Total annotations	14,129
Empty-label images	3,784 (24.75%)
Objects <1% image area	2,378 (16.83%)
Objects 1–10%	6,958 (49.25%)
Objects >10%	4,793 (33.92%)
Objects <8 px shortest side @640	92 (0.65%)
Objects <16 px shortest side @640	300 (2.12%)

The distribution showed that the dataset was not dominated by extremely tiny objects. Machette was the most small-object-sensitive class, with 37.70% of its boxes occupying less than 1% of image area. Knife followed at 22.76%, Other at 20.63%, Handgun at 12.66% and Longgun at 6.64%.

10.2 Resolution Diagnostics

Inference setting	Object TP	Object FN	Object FP	Class confusions	Image sensitivity	Image specificity
Default / standard	1	2,719	28	1	1.76%	99.94%
1280 px	72	2,635	29	14	6.45%	99.78%
1920 px	155	2,515	2,813	51	32.01%	40.25%

The resolution experiment demonstrated the central trade-off: moving from the standard operating point to 1280 px materially improved small-object recovery without creating a large false-positive increase, while direct 1920 px inference produced a sharp false-positive explosion. This was a major reason for selecting 1280 px as the practical final operating point.

10.3 Small-Object Diagnostic Set

The original 784-image Test set was also partitioned into Tiny, Small and Normal subsets based on the minimum pixel dimension of the smallest annotated object. This was a read-only diagnostic. The dataset

remained unchanged, and the subsets were used to reason about scale limitations rather than to tune repeatedly on the locked Test set.

11. Final System Architecture

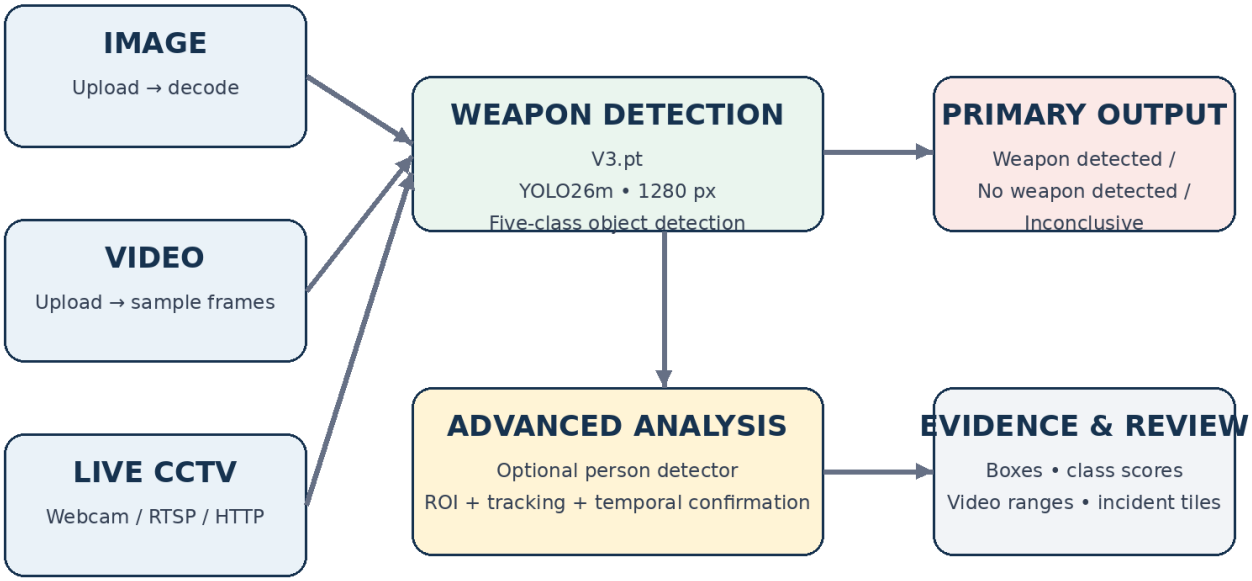


Figure 1. Final user-facing and analytical architecture. The primary path is weapon-first; person-aware analysis is optional.

11.1 Core Detection Path

The primary path is deliberately simple: input frame → V3.pt weapon detector → five-class detections → binary weapon status and evidence. This makes the weapon detector reusable across image, video and CCTV inputs without requiring a person detector for every frame.

11.2 Optional Advanced Path

The advanced path introduces person detection, person-region cropping, tracking and temporal confirmation. It is enabled only when explicitly selected. This path is useful for difficult scenes where a human subject is small, but the final aerial evaluation showed that person ROI is not automatically superior to full-frame V3 inference.

11.3 Evidence Layer

For video and CCTV, the system stores evidence that is meaningful to the user rather than exposing every inference frame. Video review groups nearby detections into ranges and lets the user seek to exact timestamps. CCTV groups consecutive detections into incident tiles and retains a peak-confidence snapshot for each incident.

12. Software Implementation

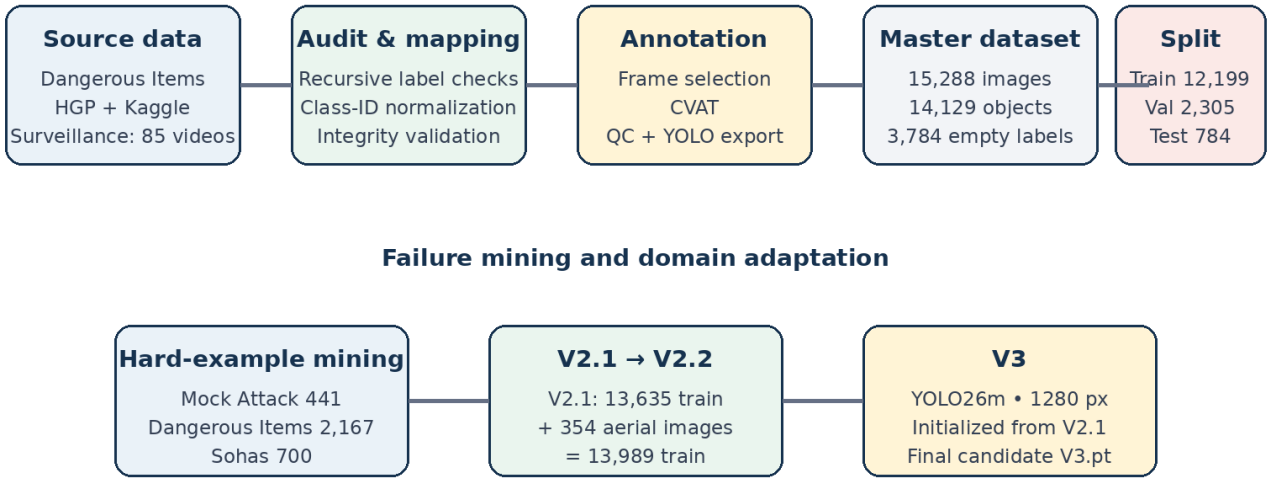


Figure 2. End-to-end data and model-development pipeline that led to the final application.

12.1 Final Project Structure

```
Wd_V3/
├── app.py
├── cctv.py
├── config.py
├── events.py
├── inference.py
├── models.py
├── pipeline.py
├── roi_processor.py
├── temporal_analysis.py
├── tracking.py
├── ui.py
├── video_processor.py
├── models/
│   ├── V2.pt
│   ├── V3.pt
│   └── yolo26n.pt
└── incidents/
```

12.2 Module Responsibilities

Module	Role
app.py	Application orchestration, Streamlit state, image/video/CCTV workflows and session controls.
ui.py	Header/sidebar, detection summaries, video player, range controls, galleries and incident presentation.

cctv.py	Camera/stream processing, latest-frame reading, incident grouping and peak-confidence snapshots.
pipeline.py	Shared image/video inference, result construction and drawing.
inference.py	YOLO model loading, prediction extraction, class validation and box helpers.
roi_processor.py	Person ROI preparation and controlled upscaling for the optional advanced path.
tracking.py	Simple IoU/center-distance person tracking across frames.
temporal_analysis.py	Temporal consistency logic; insufficient/low-resolution observations are treated as unusable.
events.py	Confirmed event creation and lifecycle management.
video_processor.py	Video metadata, upload handling and sampled-frame iteration.
models.py	Structured result objects for detections, images, videos and events.
config.py	Central model paths, thresholds, sampling settings and runtime constants.

12.3 Model Mapping

File	Meaning	Role
V3.pt	Final V3 candidate	Primary weapon detector for the frontend.
V2.pt	Retained V2.1 baseline	Comparison/fallback model; not the rejected first V2 experiment.
yolo26n.pt	Person detector	Used only under Advanced Person-Aware Analysis.

12.4 Key UI Terminology

- “Top 2 class scores” is used instead of calling raw class scores calibrated probabilities.
- “Weapon detected” and “No weapon detected” are distinct operational states built on the five-class detector.
- “Analysis inconclusive” is available as a safe interpretation when the region is too small or unusable.
- Person boxes are hidden in the default weapon-only mode and appear only in advanced analysis.

13. Image Analysis Workflow

13.1 Processing Sequence

5. User uploads an image.
6. The image is decoded and passed to the selected inference path.
7. V3.pt performs weapon detection at the configured 1280 px operating size.
8. Weapon detections are converted into class labels, confidence scores and bounding boxes.
9. The UI presents the original image, annotated image and a compact result summary.

13.2 Default versus Advanced

The default interface prioritizes direct weapon detection. The optional advanced analysis can invoke the person detector, prepare a person ROI and analyze that region. This distinction is important because the person detector is a supporting tool, not the core weapon model.

13.3 Human Review

The image result is designed to remain visually inspectable. A bounding box, class label and confidence value are shown together so that the user can distinguish a model output from a definitive fact.

14. Video Analysis Workflow

14.1 Processing

Uploaded video is analyzed through sampled frames rather than attempting to process every frame without a practical limit. The sidebar exposes a sampling control and a maximum analyzed-frame limit. Each analyzed frame retains a timestamp and frame index.

14.2 Detection Windows

Nearby detection timestamps are grouped into detection windows using the observed temporal spacing of the sampled frames. This produces meaningful ranges instead of a long list of nearly identical single-frame events.

14.3 Interactive Evidence Review

The embedded video player remains paused and provides range/timestamp shortcuts. Selecting a range stores the selected group in Streamlit session state and triggers a rerun. The analyzed VideoAnalysisResult and video bytes are also persisted in session state so that the review does not disappear when the page reruns. The selected evidence frame is shown separately and the detection gallery is kept compact.

14.4 Why This Matters

The final workflow treats the model as an evidence generator rather than an autoplay alert system. A user can move to the relevant time, inspect the source video at that location and compare it against the annotated detection.

15. Live CCTV Workflow

15.1 Source Handling

The CCTV module accepts a camera index or supported RTSP/HTTP/RTMP source. The processor uses latest-frame reading and drains stale buffered frames to keep latency low. Reconnection logic is available when the stream temporarily stops yielding frames.

15.2 Default Live Detection

The default live path applies V3.pt directly to the current full frame. If a weapon is present, the incident manager updates the current incident. Repeated detections within the configured grouping gap are treated as one incident rather than creating a tile for every frame.

15.3 Peak-Confidence Evidence

Each CCTV incident stores its peak-confidence timestamp, frame index, class and snapshot. When a stronger detection arrives within the same incident, the stored evidence snapshot is updated. This makes the incident tile representative of the strongest observed detection instead of simply the first frame.

15.4 Session Lifecycle

Action	Expected final behavior
Start monitoring	Starts a new session with empty incident tiles.
During monitoring	New detections appear as grouped incident tiles.
Stop monitoring	Current session tiles remain visible for review.
Clear current session	Current/previous session tiles are removed from the interface.
Start a new session	Old session is not restored; the new session begins clean.
Clear saved evidence	Explicitly removes stored evidence images when that control is used.

15.5 Optional Advanced CCTV Path

When advanced analysis is enabled, person detection, person tracking and temporal confirmation can be applied. This adds persistence across observations and can reduce dependence on single-frame decisions. It is intentionally not the default because the final project evidence supports a simpler full-frame weapon-first path for the general frontend.

16. Functional Verification

The final application was manually exercised during the last implementation phase. The purpose of this verification was to confirm that the working model, the evidence UI and the session state behaved as designed. Image and video behavior was preserved while the final CCTV session lifecycle was stabilized.

Test case	Expected result	Status
Image upload and inference	Image can be uploaded and annotated result is rendered.	PASS
Video upload and processing	Video metadata, analysis result and evidence are retained.	PASS
Video detection-range click	Selected time range seeks embedded video without autoplay and review remains visible.	PASS
Video evidence sizing	Selected evidence is displayed at compact review size rather than dominating the page.	PASS
CCTV start	New session starts with no previous incident tiles.	PASS
CCTV weapon detection	Detected events appear as grouped tiles with evidence.	PASS
CCTV stop	Current incident tiles remain visible after stopping.	PASS
CCTV clear current session	Previous tiles disappear only when explicitly cleared.	PASS
CCTV new session	Previous session is not restored.	PASS
Advanced person-aware mode	Person/ROI/tracking path is available only when enabled.	PASS

Verification scope

This is functional application verification, not a claim of formal certification, penetration testing, or production-grade security validation.

17. Discussion

17.1 What the Experiments Demonstrated

The project demonstrated that dataset engineering choices can affect model behavior as much as training duration. The hard-example-heavy V2 experiment had clean image/label structure but still collapsed in generalization because the overall data distribution changed too aggressively. Preserving the broad V1 distribution in V2.1 was therefore a more stable strategy.

The aerial adaptation experiment also demonstrated that targeted domain enrichment can change the operating profile of a detector. The final held-out aerial test showed a large reduction in false positives and an increase in frame-level and clip-level performance for V3 relative to V2.1 under the same full-frame evaluation procedure. At the same time, the original-domain comparison showed that this benefit was not universal across all sources.

17.2 Why Full-Frame V3 Was Selected for the Frontend

Three evidence streams informed the final pipeline: 1280 px had a better small-object trade-off than 1920 px; the final aerial test favored full-frame V3 over V3 person-ROI; and the application itself needed a simple primary path that works consistently across image, video and CCTV. These observations led to the final choice of V3.pt at full-frame 1280 px, with person-aware processing retained as an optional advanced mode.

17.3 Operational Interpretation

The detector produces five-class object-level evidence. A binary Weapon/No Weapon signal is useful as an operational summary because a class-confused weapon can still be operationally recognized as a weapon. However, binary alert performance should not be confused with object-detection mAP and should always be reported separately.

17.4 Practical Engineering Lessons

- Validate the real directory hierarchy before writing traversal scripts.
- Use complete sequence checks instead of only checking the first and last frame names.
- Never split correlated video frames randomly.
- Keep source data, normalized copies and experimental datasets separate.
- Keep benchmark metrics distinct from diagnostic counts.
- Use archive-first data transfer and local SSD extraction for large Colab jobs.
- When a checkpoint can be truly resumed, preserve the optimizer and learned state instead of resetting unnecessarily.

18. Limitations and Responsible Use

18.1 Model Limitations

- Frame-level aerial recall for V3 is 0.437 on the held-out 380-frame test, so the model misses many weapon-positive frames.
- Small and partially occluded weapons remain challenging.
- Performance varies by source domain and class; original-domain and aerial results should not be collapsed into one universal accuracy number.
- The held-out aerial test contains 380 frames from 38 clips, which is useful for controlled evaluation but is not large enough to represent every real-world aerial condition.
- Confidence thresholds and frame sampling affect operational behavior.
- CCTV grouping improves evidence presentation but does not transform the underlying single-frame detector into a formally certified surveillance system.

18.2 Dataset Limitations

The source datasets differ in visual style, annotation conventions and semantic coverage. The Other class is intentionally broad. Some historical sources also contain noisy or broad labels. These characteristics were documented rather than hidden, but they remain part of the system limitations.

18.3 Responsible Use

The final system should be presented as a research/portfolio proof of concept. It should not be treated as a sole decision-maker for high-consequence security actions. Human verification remains appropriate, especially when detections are small, occluded, ambiguous or otherwise difficult to interpret.

19. Conclusion and Project Closure

This project evolved from a raw multi-source weapon-detection dataset into a complete, tested application with a controlled model-development history. The work addressed the full lifecycle: source auditing, label normalization, surveillance-frame extraction, temporal leakage prevention, redundancy-aware frame selection, CVAT annotation, YOLO export validation, master dataset construction, baseline training, hard-example mining, aerial adaptation, controlled evaluation, and frontend integration.

The final development path preserved the lessons from the failed first V2 rather than hiding them. V2.1 restored the broad training distribution and became the retained baseline. V2.2 added a targeted 354-image aerial adaptation set without contaminating the held-out aerial clips. V3 was then trained at 1280 px and evaluated on both the original locked Test set and a separate held-out aerial benchmark.

The final frontend is intentionally weapon-first. V3.pt is the primary detector, while yolo26n.pt exists only for optional person-aware analysis. Image, video and live CCTV modes now form one consistent application. Video review provides interactive, paused evidence inspection. CCTV provides grouped incident tiles, peak-confidence snapshots and a predictable session lifecycle in which stopping preserves the session, clearing explicitly removes it, and a new session starts clean.

Project closure
Model development is frozen. The final deliverable is the V3.pt-based Weapon Detection application together with its validated datasets, evaluation evidence, supporting modules and this complete project report.

References

10. Ultralytics. Ultralytics YOLO documentation and object-detection framework. Used for model training, inference and YOLO dataset handling.
11. OpenCV. Open Source Computer Vision Library documentation. Used for image decoding, video capture, frame handling and evidence rendering.
12. Streamlit. Streamlit documentation. Used for the interactive application frontend and session-state-driven user interface.
13. CVAT. Computer Vision Annotation Tool documentation. Used for local image annotation, tracking and YOLO export.
14. PyTorch. PyTorch documentation. Used as the underlying deep-learning framework and GPU runtime for training and inference.
15. Project Continuity, Final Development & Evaluation Record (updated 21 September 2026). Internal project record supplying dataset, experiment, evaluation and artifact history.
16. Final project implementation files: app.py, cctv.py, ui.py, pipeline.py, inference.py, models.py, roi_processor.py, tracking.py, temporal_analysis.py, events.py and video_processor.py.

Appendix A. Final Project Counts

Artifact / stage	Count / state
Surveillance source videos	85
Surveillance extracted frames	3,197
Final annotation candidates	890
Verified master images	15,288
Verified master annotations	14,129
Empty-label images	3,784
Final Train / Validation / Test	12,199 / 2,305 / 784
V2.1 Train images	13,635
Aerial adaptation images	354
V2.2 Train images	13,989
Held-out aerial test	380 frames / 38 clips
Final V3 checkpoint	V3.pt

Appendix B. Final Class Mapping

ID	Name	Primary examples
0	Knife	Knife
1	Machette	Machete
2	Handgun	Pistol / handgun
3	Longgun	Rifle / shotgun / long firearm
4	Other	Other in-scope dangerous item

Appendix C. Final Model Evidence

Evaluation	Model	Precision	Recall	mAP50	mAP50-95 / F1

Historical locked original Test	V1 Epoch-46	0.7211	0.5467	0.5655	0.4111 mAP50-95
Original Test, 1280 px	V2.1	0.678	0.628	0.601	0.413 mAP50-95
Original Test, 1280 px	V3	0.658	0.544	0.538	0.356 mAP50-95
Held-out aerial, frame level	V3 full-frame	0.830	0.437	-	0.572 F1
Held-out aerial, clip level	V3 full-frame	0.765	0.684	-	0.722 F1

Appendix D. Final Functional Behavior Checklist

- Image analysis works with V3.pt and optional advanced person-aware analysis.
- Video analysis retains results across Streamlit reruns and provides grouped detection ranges.
- Video seek controls do not autoplay and the selected evidence remains visible.
- Live CCTV uses latest-frame capture and groups continuous detections into incidents.
- CCTV stop preserves the current session for review.
- Clear Current Session removes the previously detected incident tiles.
- A newly started CCTV session does not restore the prior session.
- Incident evidence uses peak-confidence snapshots when a stronger detection is observed.

Appendix E. Final Artifact Recovery Map

Artifact	Final role
Wd_V3/models/V3.pt	Final frontend weapon detector
Wd_V3/models/V2.pt	V2.1 comparison/fallback model
Wd_V3/models/yolo26n.pt	Optional person detector
Wd_V3/app.py	Application orchestration and session handling
Wd_V3/ui.py	User interface and video/CCTV evidence presentation
Wd_V3/cctv.py	Live stream capture and incident grouping
Final_Dataset_Split	Verified base Train/Validation/Test dataset
V2_Training_Dataset_V2.2.zip	Final V2.2 training archive used for V3

Aerial_Final_Evaluation/locked_test	Held-out aerial evaluation frames
Final evaluation outputs	Original-domain and aerial comparison artifacts

Appendix F. Project Closure Record

Final implementation status: Complete as of 27 September 2026. No further model training is required for the finalized project scope. Additional experiments may be considered in a future research cycle, but they are outside the closed implementation documented in this report.